

Exam Date & Time: 07-Dec-2023 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER / DECEMBER 2023

SUBJECT: CSE 2123- DIGITAL SYSTEM DESIGN

DIGITAL SYSTEM DESIGN [CSE 2123]

Marks: 50

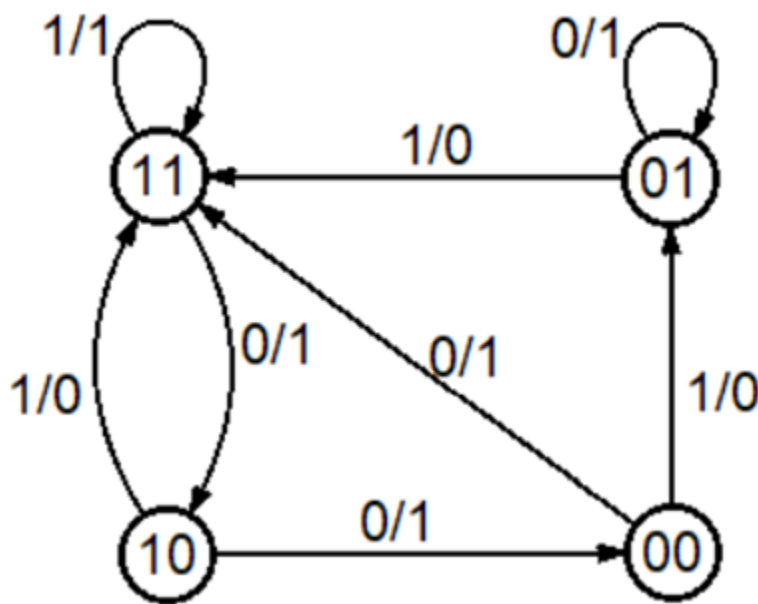
Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

- 1A) For the following state diagram, derive the expressions required for the synchronous sequential circuit design using JK flip-flops.



(5)

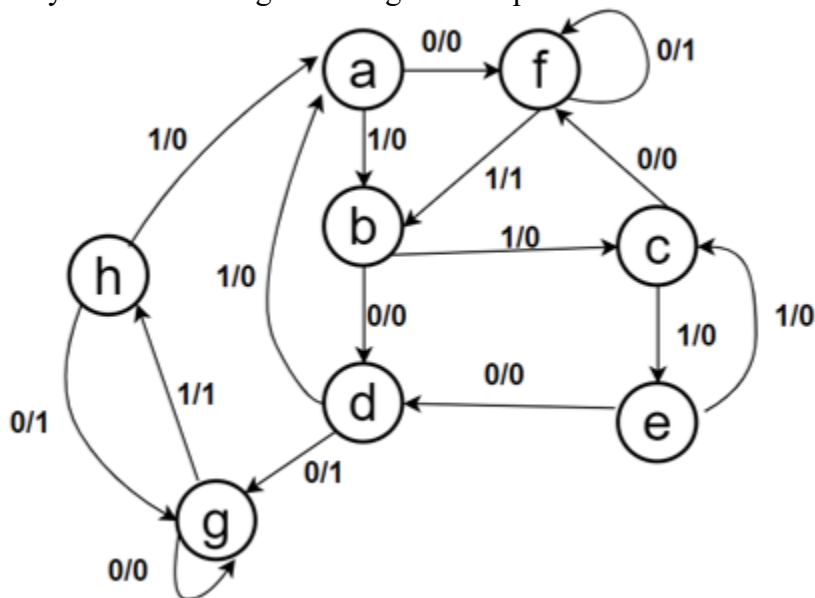
- 1B) Make use of functional decomposition to find the minimum-cost circuit for the function $f(x_1, x_2, x_3, x_4) = \sum m(4, 7, 9, 10, 12, 13, 14, 15)$. Assume that the input variables are available in uncomplemented form only. (3)
- 1C) Demonstrate SR latch (basic Flip Flop circuit) using NAND gates with its function table. (2)
- 2A) It is required to develop a system to swap the contents of two n-bit registers. Assume that a temporary register is available to assist swapping. Adopt state table approach to design control circuit. Control circuit has two inputs clock and w. The swapping process is initiated when w=1, and on completion the control circuit asserts a signal Done to 1. (5)
- Derive the expression for the outputs of control circuit.
 - Develop bus architecture of the system based on tri-state buffer.

- 2B) A combinational circuit receives two single-bit inputs A and B and produces a single-bit output $Y = F1 \oplus F2$. The Boolean functions F1 and F2 are as given below and are implemented using a decoder and OR gates. Design a minimal circuit for Y using only one decoder, one 2:1 multiplexer, and OR gates. (3)
- $F1(A, B) = \sum m(2, 3)$
 $F2(A, B) = \sum m(0, 1, 3)$

- 2C) Construct JK Flip Flop from D Flip Flop (2)

- 3A) Design the synchronous counter with the following repeated binary sequence: 0, 1, 2, 4, 6. Use D flip-flops. Check if it is self-correcting or not. (5)

- 3B) Analyze the following state diagram and perform the state reduction. (3)



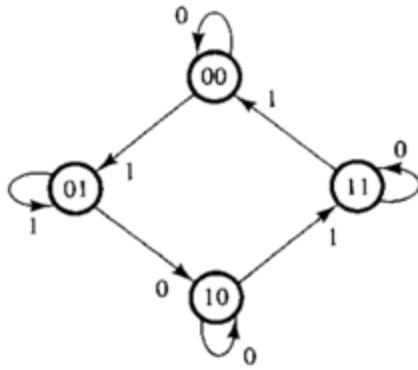
- 3C) Distinguish between Moore and Mealy models of sequential circuit design. (2)

- 4A) Since a computer has many registers, paths must be provided to transfer information from one register to another. An efficient scheme for transferring the information between registers in a multiple-register configuration is a common bus system. Design a 4-bit common bus system for 4-registers of 4-bits each using multiplexers. (5)

- 4B) Design 3-bit Johnson counter by implementing the following steps. (3)
- Draw the state-transition diagram.
 - Construct state table using T flip-flops.
 - Design the Counter.

- 4C) Design NAND gate using CMOS technology. (2)

- 5A) Draw the state table and design the synchronous sequential circuit for the following state diagram using JK FF: (4)



5B) Design a combinational unit that can act as adder and subtractor for two 4-bit numbers. Provide suitable explanation. (4)

5C) Design a 4-bit ring counter using D flip-flops. (2)

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